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Documentation on LVM and SELinux

Project Title: *Scalable Enterprise Storage Deployment & Hardening*

Objective: The goal of this project was to implement a dynamic storage solution using LVM to ensure the infrastructure can handle future data growth without downtime. Simultaneously, SELinux configurations were managed to balance system security with application availability during the initial deployment phase.

1.LVM:

1. Disk Partitioning (fdisk)

Prepare the physical disk and set the partition type to LVM:

- **fdisk /dev/sdb** — Open disk utility.
- n — Create new partitions.
- t — Change partition type to **8e** (Linux LVM).
- w — Write changes and exit.

```
[root@localhost ~]# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.32.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0xeb598134.

Command (m for help): clear
DOS Compatibility flag is set (DEPRECATED!)

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (63-10485759, default 64): 64
Last sector, +sectors or +size{K,M,G,T,P} (64-10485759, default 10485759): +3G

Created a new partition 1 of type 'Linux' and of size 3 GiB.

Command (m for help): t
```

- **Setting Type:** Enter t to change the partition type to **8e** (Linux LVM).

```
 9  AIX bootable    4f  QNX4.x 3rd part 93  Amoeba          e1  DOS access
 a  OS/2 Boot Manag 50  OnTrack DM      94  Amoeba BBT      e3  DOS R/O
 b  W95 FAT32       51  OnTrack DM6 Aux 9f  BSD/OS          e4  SpeedStor
 c  W95 FAT32 (LBA) 52  CP/M            a0  IBM Thinkpad hi ea  Rufus alignment
 e  W95 FAT16 (LBA) 53  OnTrack DM6 Aux a5  FreeBSD         eb  BeOS fs
 f  W95 Ext'd (LBA) 54  OnTrackDM6      a6  OpenBSD         ee  GPT
10  OPUS            55  EZ-Drive        a7  NeXTSTEP        ef  EFI (FAT-12/16/
11  Hidden FAT12    56  Golden Bow      a8  Darwin UFS      f0  Linux/PA-RISC b
12  Compaq diagnost 5c  Priam Edisk     a9  NetBSD          f1  SpeedStor
14  Hidden FAT16 <3 61  SpeedStor       ab  Darwin boot     f4  SpeedStor
16  Hidden FAT16    63  GNU HURD or Sys af  HFS / HFS+      f2  DOS secondary
17  Hidden HPFS/NTF 64  Novell Netware  b7  BSDI fs         fb  VMware VMFS
18  AST SmartSleep  65  Novell Netware  b8  BSDI swap       fc  VMware VMKCORE
1b  Hidden W95 FAT3 70  DiskSecure Mult bb  Boot Wizard hid fd  Linux raid auto
1c  Hidden W95 FAT3 75  PC/IX           bc  Acronis FAT32 L fe  LANstep
1e  Hidden W95 FAT1 80  Old Minix       be  Solaris boot    ff  BBT
Hex code (type L to list all codes): 8e
Changed type of partition 'Linux' to 'Linux LVM'.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

```
[root@localhost ~]# lsblk
NAME          MAJ:MIN RM SIZE RO TYPE MOUNTPOINT
sda             8:0    0  30G  0 disk
├sda1           8:1    0   1G  0 part /boot
└sda2           8:2    0  29G  0 part
  ├rhel-root 253:0    0  26G  0 lvm  /
  └rhel-swap 253:1    0   3G  0 lvm  [SWAP]
sdb             8:16   0   5G  0 disk
├sdb1           8:17   0   3G  0 part
└sdb2           8:18   0   2G  0 part
```

```
[root@localhost ~]# df -h
Filesystem                Size  Used Avail Use% Mounted on
devtmpfs                  1.4G     0  1.4G   0% /dev
tmpfs                     1.4G     0  1.4G   0% /dev/shm
tmpfs                     1.4G   18M  1.4G   2% /run
tmpfs                     1.4G     0  1.4G   0% /sys/fs/cgroup
/dev/mapper/rhel-root      26G  4.3G   22G  17% /
/dev/sda1                1014M  228M  787M  23% /boot
tmpfs                     278M   16K  278M   1% /run/user/42
tmpfs                     278M   20K  278M   1% /run/user/0
/dev/mapper/NewVG-OurLVM  1.9G  5.9M  1.8G   1% /mnt/bashar data
[root@localhost ~]#
```

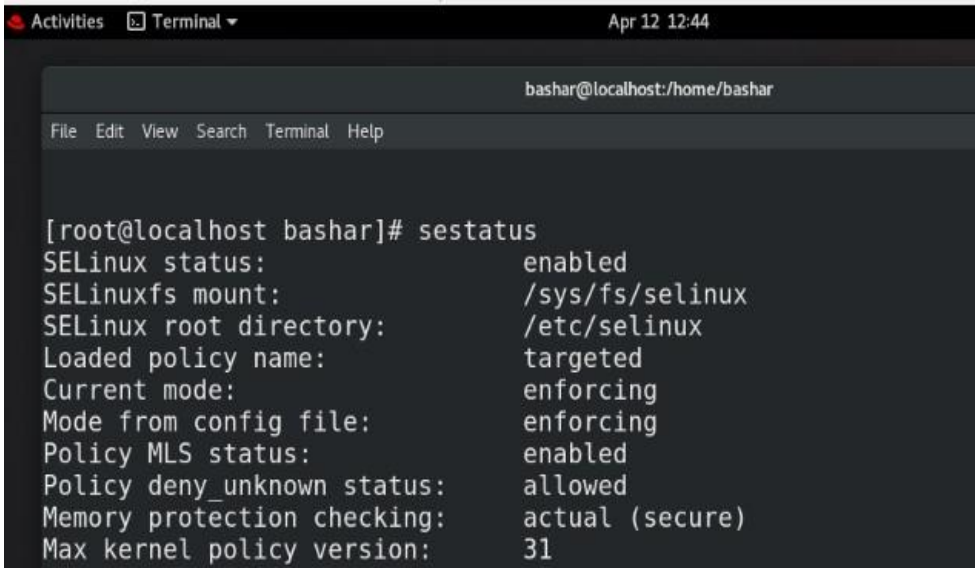
- **Verification:**
  - **lsblk** is used to verify the new partitions (e.g., sdb1 and sdb2).
  - **df -h** confirms the successfully mounted LVM volume (e.g., /dev/mapper/NewVG-Our LVM mounted on **/mnt/bashar data**).

2. Selinux:

a) To switch into permissive mode for command: **setenforce 0**

```
[root@localhost bashar]# setenforce 0
[root@localhost bashar]# setenforce 0
[root@localhost bashar]# sestatus
SELinux status:                 enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:             targeted
Current mode:                   permissive
Mode from config file:          enforcing
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Memory protection checking:     actual (secure)
Max kernel policy version:      31
[root@localhost bashar]#
```

b) For see the Status: (**sestatus/ getenforce**)



Conclusion: This workflow demonstrates a fundamental understanding of Linux system administration. Transitioning from physical disks to **Logical Volumes** is an industry standard for maintaining high availability. While setenforce 0 is an excellent diagnostic tool, the final phase of this project would involve re-enabling **Enforcing** mode after creating the necessary policy exceptions, ensuring the server is both flexible and battle-hardened against external threats.